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BACK UP AID FOR RIDE-ON EQUIPMENT

Abstract

A back up system for ride-on equipment includes a plurality of back up sensors configured to sense an object within a back up region of the ride-on equipment, a deterrence device configured to emit a sound, a display configured to display the sensed object, and a controller configured to: determine whether the sensed object is within the back up region based on feedback from the plurality of back up sensors, and activate, in response to determining that the object is within the back up region, the deterrence device and the display.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/561,546, filed Mar. 5, 2024, the entire content of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to back up aids for ride-on equipment, and, more specifically, to back up aids for detecting objects or beings behind the ride-on equipment and alerting an operator.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a back up system for ride-on equipment including: a plurality of back up sensors configured to sense an object within a back up region of the ride-on equipment; a deterrence device configured to emit a sound; a display configured to display the sensed object; and a controller configured to: determine whether the sensed object is within the back up region based on feedback from the plurality of back up sensors, and activate, in response to determining that the object is within the back up region, the deterrence device and the display.

[0004] In some aspects, the techniques described herein relate to a back up system for ride-on equipment including: a sensor configured to sense an object within a back up region of the ride-on equipment; a deterrence device configured to emit a sound; an indicator; and a controller configured to: determine whether the sensed object is within the back up region based on feedback from the sensor, and activate, in response to determining that the object is within the back up region, the deterrence device and the indicator.

[0005] In some aspects, the techniques described herein relate to a back up system for ride-on equipment including: a plurality of back up sensors configured to sense an object within a back up region of the ride-on equipment the plurality of back up sensors including at least one infrared (IR) sensor; a deterrence device configured to emit sound waves in a frequency range from 17,000 Hz to 19,000 Hz; and a controller configured to: determine whether the sensed object is within the back up region based on feedback from the plurality of back up sensors, determine whether the sensed object is a living being based on feedback from the IR sensor. activate, in response to determining that the object is within the back up region and that the object is a living being, the deterrence device.

[0006] Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a schematic illustration of ride-on equipment including a back up system embodying aspects of the present disclosure.

[0008] FIG. **2** is a schematic illustration of the ride-on equipment of FIG. **1**, illustrating detection of an object by back up system.

[0009] FIG. **3** is schematic illustration of a deterrence device of the ride-on equipment of FIG. **1**, according to some aspects and examples.

[0010] FIG. **4** is a schematic illustration of a display of the ride-on equipment of FIG. **1**, according to some aspects and examples.

[0011] FIG. **5** is a schematic illustration of the display of the ride-on equipment of FIG. **1**, indicating a location of an objected detected by the back up system.

[0012] FIG. **6** is a schematic illustration of the display of the ride-on equipment of FIG. **1**, indicating a location of a living being detected by the back up system.
[0013] FIG. **7** is a block diagram of a controller for the ride-on equipment of FIG. **1**, according to some aspects and examples.
[0014] FIG. **8** is a flow chart of a first method of operation of the back up system of the ride-on equipment of FIG. **1**.
[0015] FIG. **9** is a flow chart of a second method of operation of the back up system of the ride-on equipment of FIG. **1**.
[0016] FIG. **10** is a plan view of a deterrence device usable with the ride-on equipment of FIG. **1**.

DETAILED DESCRIPTION

[0017] Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.
[0018] FIG. **1** illustrates an example of ride-on equipment **100** in the form of a lawn mower. The equipment **100** may include a rotating blade for cutting a lawn to a desired height. In other embodiments, the equipment **100** may additionally or alternatively include a tool for sweeping debris, vacuuming debris, clearing debris, collecting debris, moving debris, etc. Debris may include plants (such as grass, leaves, flowers, stems, weeds, twigs, branches, etc., and clippings thereof), dust, dirt, jobsite debris, snow, or any other outdoor lawn or landscape debris. In the illustrated embodiment, the equipment **100** is a sitting ride-on lawn mower. In other embodiments, the equipment **100** may be a standing ride-on lawn mower, zero-turn mower, etc. The equipment **100** may be autonomous, semi-autonomous, or not autonomous. In yet other embodiments, the equipment **100** may include any other ride-on equipment configured to move while carrying an operator, such as offroad vehicles, powersport equipment, ATVs, UTVs, forklifts, maintenance, or landscaping equipment.
[0019] The illustrated equipment **100** includes a right front wheel **102**, a left front wheel **104**, a right rear drive wheel **106**, a left rear drive wheel **108**, a chassis or frame **110**, an operator platform or seat **112** coupled to the frame **110** and configured to support an operator driving the equipment **100**, and a control unit **114**, which may include levers, joysticks, a steering wheel, or the like used to control operation of the equipment **100**. The equipment **100** is drivable in a forward direction and a rearward or reverse direction. The equipment **100** further includes a back up system **122** configured to detect objects and living beings (i.e., people and animals) proximate the equipment **100** and provide information/warnings regarding detected objects and living beings to an operator of the equipment **100**. For example, as described in greater detail below, the illustrated back up system **122** includes one or more back up sensors **126** configured for detecting the presence and location of objects and living beings within a field of view of the back up sensors **126** and an indicator, such as a display **134**, for displaying the sensed objects and living beings to the operator. The illustrated back up system **122** may additionally or alternatively include a deterrence device **130** configured to deter living beings from a path of the equipment **100**, as described in greater detail below.
[0020] With reference to FIGS. **1** and **2**, the frame **110** includes a rear portion **118** configured for mounting the back up sensors **126**. The back up sensors **126** are configured to detect objects or living beings that come within a projected back up region **138** of the equipment **100**. In some embodiments, the back up region **138** is an approximately 16 square meter region behind the equipment **100**. For example, the back up region **138** may be approximately 4 meters long and 4 meters wide. In alternate embodiments, the back up region **138** may vary in size (e.g., depending on the size and speed of the equipment **100**).
[0021] In the illustrated embodiment, the back up sensors **126** include one or more time of flight

(ToF) sensors able to measure the distance between the sensor **126** and an object or being by emitting energy and then detecting the energy reflected back by the object or being. For example, the back up sensors **126** may include ultrasonic sensors, infrared sensors, mmWave sensors, and the like. The illustrated back up sensors **126** include a first sensor **142**, which is an ultrasonic sensor **142** in the illustrated embodiment, and second sensor **146**, which is an infrared (IR) sensor **146** in the illustrated embodiment. The ultrasonic sensor **142** is configured to measure a distance from the rear portion **118** of the equipment **100** to an object **128** within the back up region **138** (FIG. **2**). In particular, the ultrasonic sensor **142** includes one or more ultrasonic emitters and one or more receivers that measure an echo (i.e., the return of the sound waves emitted by the one or more ultrasonic emitters) and convert the echo to a sensor signal (e.g., a voltage/current) indicative of a size and position of the object relative to the ultrasonic sensor **142**. The sensor signal is then received and processed by a controller **200** (FIG. **7**) of the equipment **100**, described in greater detail below.

[0022] The IR sensor **146** is configured to sense the temperature, size, and/or position of the object **128** within the back up region. In the illustrated embodiment, the IR sensor **146** includes one or more IR light detectors. The IR light detectors may receive IR light emitted by the object **128**, which can then be used to determine a temperature of the object **128** and identify whether the temperature of the object **128** is consistent with the temperature of a living being. The IR sensor **146** may also includes one or more IR emitters, which may direct an IR beam having a beam angle between about 5 and 165 degrees. In the illustrated embodiment, the beam angle is between about 15 and 90 degrees. The detector(s) of the IR sensor **146** may sense light reflected back from the object **128** and provide a signal indicative of the size and position of the object **128** relative to the IR sensor **146**. In some embodiments, the IR sensor **146** may be used in combination with the ultrasonic sensor **142** (or another ToF sensor) to determine the size and position of the object **128** and whether the object **128** is a living being or an inanimate object.

[0023] In some embodiments, the IR sensor **146** may be placed within the frame **110**. In such an embodiment, at least a portion of the frame **110** may be transmissive of electromagnetic radiation, specifically transmissive of infrared radiation, so that the IR sensor **146** is placed within the frame while the IR light beam travels through the transmissive frame **110** for the IR sensor **146** to sense the living being. In other embodiments, the IR sensor **146** and the ultrasonic sensor **142** may be provided together in a sensor housing coupled to the frame **110** in any suitable manner.

[0024] In some embodiments, the back up sensors **126** may additionally or alternatively include a mmWave radar emitter and sensor, which may be used by the controller **200** to detect objects, their relative position, and determine whether the detected objects are living beings.

[0025] In some embodiments, the back up sensors **126** may additionally or alternatively include an optical sensor to distinguish living beings from surrounding plants and objects. The back up sensors **126** may additionally or alternatively include a light sensor. The light sensor detects light of the surrounding area. The light sensor may be implemented using a phototransistor, a photodiode, a photonic integrated circuit, or the like.

[0026] In some embodiments, the back up sensors **126** may additionally or alternatively include a vision system with one or more cameras able to detect and recognize objects and beings behind the equipment **100**. For example, in such embodiments, the second sensor **146** may include a camera, and the controller **200** may use image processing and artificial intelligence to analyze images from the camera and determine whether a living being is positioned within the field of view of the second sensor **146**. In some embodiments, the camera may be used both to determine a relative position of the detected object and whether the detected object is a living being. In some embodiments, the first sensor **142** (e.g., a ToF sensor such as an ultrasonic sensor) may be used by the controller **200** for position and distance determinations.

[0027] With reference to FIG. **3**, the deterrence device **130** of the back up system **122** may be a sound emitting device configured to deter children or animals within a distance range of the equipment **100**. The range may include an approximately 10-meter radius surrounding the equipment **100**. In other embodiments, the radius may be any radius between 0-meters and 100-meters, for example. The deterrence device **130** is configured to emit sound waves that are of a high frequency, such that only children or animals hear the sound. The frequency of the sound waves is a frequency that is unpleasant to hear such that neither the animal nor the child wishes to be near the source. Specifically, the frequency of the sound emitted by the deterrence device **130** may range from 17,000 Hz to 25,0000 Hz which is a frequency range that is known to effectively deter humans and animals. In some embodiments, the frequency may range from 17,000 Hz to 19,000 Hz, or about 18,000 Hz in some embodiments. The amplitude of the sound waves is controlled such that the sound emitted from the deterrence device **130** may be heard while not causing any permanent hearing damage. In the present embodiment, the deterrence device **130** is mounted to a central portion of the frame **110** to maintain a consistent deterrence range. In alternate embodiments, the equipment **100** may include a plurality of deterrence devices **130** spaced around the frame **110** to extend the range. In the present embodiment, the deterrence device **130** emits a sound when a living being is sensed by the back up sensors **126**. The deterrence device **130** continues emitting the sound until the back up sensors **126** no longer sense the living being. In some embodiments, the deterrence device **130** may continuously emit a sound (e.g., when the equipment **100** is operating).

[0028] The deterrence device **130** may include a variety of different configurations for emitting the deterrence sound. In some embodiments, the deterrence device **130** includes one or more speakers. In other embodiments, the deterrence device **130** includes one or more piezo buzzers. In yet other embodiments, the deterrence device **130** includes one or more sirens.

[0029] For example, with reference to FIG. **10**, a siren **304** that may be incorporated into the deterrence device **130** includes a stator **308** surrounding a rotor or fan **312**. The stator **308** includes a first plurality of fins **316** extending in a radial direction and spaced apart around the circumference of the stator **308**. The fan **312** includes a second plurality of fins **320** extending in the radial direction and spaced apart around the circumference of the fan **312**. The fan **312** also includes a hub **324** attached to a drive, such as an electric motor (not shown). When the fan **312** is driven to rotate within the stator **308**, a tone is produced by the siren **304**. In the illustrated embodiment, the first plurality of fins **316** and the second plurality of fins **320** are the same in number. In some embodiments, the first and second pluralities of fins **316**, **320** may each include between 90-150 fins, or between 110-130 fins in other embodiments. In the illustrated embodiment, the first and second pluralities of fins **316**, **320** each include 120 fins. The number of fins **316**, **320** and the rotational speed of the fan **312** control the frequency of the tone emitted by the siren **304**. In some embodiments, the fan **312** may rotate between 7,000 and 11,000 RPM, and the fan **312** is configured to rotate at about 9,000 RPM in the illustrated embodiment, which provides the desired tone frequency for the deterrence device **130**.

[0030] Referring to FIGS. **1-3**, the display **134** may be positioned adjacent the control unit **114** such that the operator of the equipment **100** may view the display **134** while operating the equipment **100**. For example, the display **134** may include a display screen integrated into the equipment **100**. In other embodiments, the display **134** may be an indicator located on operating handles (e.g., the control unit **114**) of the equipment **100**. The operating handles may be used to control forward and reverse operation and/or steering of the equipment **100**. In such embodiments, the indicator may include LED light strips applied to or incorporated into the operating handles.

[0031] The display **134** is configured to warn the operator of the object **128** in the back up region **138** so that the operator may avoid the object **128**. FIGS. **4-6** illustrate exemplary graphics that may be displayed by the display **134** to convey information to the operator of the equipment **100**. The illustrated display **134** shows a schematic of the equipment **100** and the back up region **138**. The schematic of the back up region **138** may include a grid of symbols or shapes (e.g., eight rectangles **150** organized into two columns, or any other suitable arrangement). The rectangles **150** may be color coded based on a relative distance away from the equipment **100** and may be used to identify the location of the sensed object.

[0032] For example, FIG. **4** represents a “No Object Sensed” display configuration **162**, which depicts the eight rectangles **150** having a uniform size and shape. Thus, in response to no object being sensed, all eight rectangles **150** are the same shape and size to signify to the operator that an object or living being has not been sensed by the back up sensors **126**. FIG. **5** represents an “Object Sensed” display configuration **166**, which depicts a larger rectangle or highlighted region **154** overlaid on top of one or several of the rectangles **150**. Thus, in response to the back up sensors **126** sensing an object **128** within the back up region **138**, the larger rectangle **154** is overlaid on top of whichever of the eight rectangles **150** is closest to the location of the object **128** to alert the operator both that an object was sensed and the general location of the sensed object **128**. In the case that the object **128** is located between several rectangles **150** or the object **128** is larger than one of the rectangles **150**, then several rectangles **150** may have a larger rectangle overlaid.

[0033] With reference to FIG. **6**, the display **134** may also have a warning display configuration **170** if the back up sensors **126** detect a living being within the back up region **138**. In the illustrated embodiment, the display may show a person shape **174** between the rectangles **150** in response to the back up sensors **126** sensing a living being within the back up region **138**. In some embodiments, the person shape **174** may be positioned over the rectangles **150** at a location corresponding with the sensed position of the living being.

[0034] The controller **200** for the equipment **100** will now be described with reference to FIG. **7**. The controller **200** is electrically and/or communicatively connected to a variety of modules or components of the equipment **100**. For example, the illustrated controller **200** may be connected to the deterrence device **130**, a power input unit **260**, the back up sensors **126**, and the display **134**. In some embodiments, the controller **200** may be housed together with the display **134**; however, the controller **200** may alternatively be located elsewhere on or within the equipment **100**.

[0035] The controller **200** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller **200** and/or equipment **100**. For example, the controller **200** may include, among other things, a processing unit **205** (e.g., a microprocessor, an electronic processor, an electronic controller, a microcontroller, or another suitable programmable device), a memory **225**, input units **230**, and output units **235**. The processing unit **205** may include, among other things, a control unit **210**, an arithmetic logic unit (“ALU”) **215**, and a plurality of registers **220** implemented using a known computer architecture (e.g., a modified Harvard architecture, a von Neumann architecture, etc.). The processing unit **205**, the memory **225**, the input units **230**, and the output units **235**, as well as the various modules connected to the controller **200**, may be connected by one or more control and/or data buses.

[0036] The memory **225** is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as a ROM, a RAM (e.g., DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The processing unit **205** is connected to the memory **225** and executes software instructions that are capable of being stored in a RAM of the memory **225** (e.g., during execution), a ROM of the memory **225** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in the implementation of the equipment **100** can be stored in the memory **225** of the controller **200**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The controller **200** is configured to retrieve from the memory **225** and execute, among other things, instructions related to the control processes and methods described herein. In other embodiments, the controller **200** includes additional, fewer, or different components.

[0037] The signals from the back up sensors **126** may pass from the input units **230**, along control lines (buses) **240**, and are processed by the processing unit **205**, which then controls the information displayed on the display **134** via the output units **235**. The deterrence device **130** is also connected to the controller **200** (via the output units **235**) and receives control signals from the controller **200** to turn on and off or otherwise convey information based on different states of the equipment **100**. Thus, the controller **200** is programmed to control the deterrence device **130** and the display **134** based on feedback from the back up sensors **126**. The controller **200** is further connected to a power input unit **260** to regulate or control the power to and/or from the controller **200**.

[0038] FIGS. **8** illustrates an exemplary method for activating the deterrence device **130** and the display **134**. The method may be performed automatically by the controller **200** in some embodiments. The steps of the method are described in an iterative manner for descriptive purposes. Various steps described herein with respect to the method are capable of being executed simultaneously, in parallel, or in an order that differs from the illustrated serial and iterative manner of execution.

[0039] At step **805**, the controller **200** determines whether an object **128** is in range of the equipment **100**. The controller **200** determines if an object **128** is in range of the equipment **100** via the first sensor **142**, which senses if the object **128** is within the back up region **138**. It should be understood that “in range” means within the back up region **138**. In some embodiments, step **805** may include using the second sensor **146** in addition to or in alternative to the first sensor **142**. NO goes to step **805**, and the display **134** may remain in the “No Object Sensed” configuration **162** shown in FIG. **4**. YES goes to step **810**. At step **810**, in response to determining that the object **128** is in range of the equipment **100**, the controller **200** activates the deterrence device **130**. Step **810** may additionally or alternatively include illuminating lights (turn signal lights, head lights, brake lights, warning lights, or other indicator lights) in a blinking pattern to draw attention.

[0040] Step **810** may additionally or alternatively include activating the “Object Sensed” configuration **166** of the display **134**, illustrated in FIG. **5**. At step **815**, the controller **200** activates the second sensor **146**. In some embodiments, other sensors may be activated in addition to the second sensor **146**. At step **820**, the controller **200** determines whether the object **128** is sensed by the second sensor **146**. In response to the object **128** being sensed by the second sensor **146**, the controller **200** determines whether a living being is within the back up region **138** (e.g., based on a temperature of the object **128** when the second sensor **146** is an IR sensor, or based on interpreting an image of the object **128** when the second sensor **146** is a camera). NO goes to step **805**. YES goes to step **825**. At step **825**, in response to the object **128** being sensed and determined to be a living being, the controller **200** activates the warning display configuration **170** to indicate to the operator that a living being is within the back up region **138**. Once the warning display configuration **170** has been activated, the method returns to step **805**. Steps **805**, **810**, **815**, **820**, **825** repeat until the object **128** is no longer in range of the equipment **100** (NO at step **805**). In some embodiments, the controller **200** may activate the deterrence device **130** only if the controller **200** has determined that the object **128** is a living being (i.e., step **810** may occur after step **820** and together with step **825**).

[0041] As another example, FIG. **9** provides a second method for activating the back up system **122**. The method may be performed automatically by the controller **200**. The steps of the method are described in an iterative manner for descriptive purposes. Various steps described herein with respect to the method are capable of being executed simultaneously, in parallel, or in an order that differs from the illustrated serial and iterative manner of execution.

[0042] At step **905**, the controller **200** determines whether an object **128** is in range of the equipment **100**. The controller **200** determines if an object **128** is in range of the equipment **100** via the first sensor **142** which senses if an object **128** is within the back up region **138**. In some embodiments, step **905** may include using the second sensor **146** in addition to or in alternative to the first sensor **142**. In some embodiments, other sensors may be activated in addition to the second sensor **146**. NO goes to step **905**. YES goes to step **910**. At step **910**, in response to determining that an object **128** is in range of the equipment **100**, the controller **200** activates the deterrence device **130**. At step **915**, the controller **200** activates the warning configuration **170** of the display **134**. Once the warning configuration **170** has been activated, the method returns to step **905**. Steps **905**, **910**, **915** repeat until the object **128** is no longer in range of the equipment **100** (NO at step **905**).

[0043] Thus, embodiments described herein provide, among other things, systems, methods, and devices for back up sensing and sound deterrents for lawn equipment. Various features and advantages are set forth in the following claims.

Claims

1. A back up system for ride-on equipment comprising: a plurality of back up sensors configured to sense an object within a back up region of the ride-on equipment; a deterrence device configured to emit a sound; an indicator; and a controller configured to: determine whether the sensed object is within the back up region based on feedback from the plurality of back up sensors, and activate, in response to determining that the object is within the back up region, the deterrence device and the indicator.

2. The back up system of claim 1, wherein at least one of the plurality of back up sensors is a time of flight sensor.

3. The back up system of claim 1, wherein at least one of the plurality of back up sensors is a camera.

4. The back up system of claim 3, wherein the controller is configured to determine whether the sensed object is a living being based on feedback from the camera.

5. The back up system of claim 4, wherein the controller is configured to activate the deterrence device only if the controller determines that the sensed object is a living being.

6. The back up system of claim 4, wherein the indicator is a display, and wherein the controller is configured to generate a warning on the display to indicate that a living being is within the back up region in response to determining that the sensed object is a living being.

7. The back up system of claim 5, wherein the deterrence device is configured to emit sound waves in a frequency range between 17,000 Hz and 19,000 Hz.

8. The back up system of claim 7, wherein the deterrence device includes a piezo buzzer.

9. The back up system of claim 7, wherein the deterrence device includes a siren having a rotatable fan and a stator surrounding the fan, the stator including a first plurality of fins and the fan including a second plurality of fins.

10. The back up system of claim 9, wherein the first plurality of fins and the second plurality of fins include the same number of fins.

11. The back up system of claim 10, wherein the number of fins is between 90 and 150 fins.

12. The back up system of claim 11, wherein the fan is rotatable relative to the stator at a speed between 7,000 RPM and 11,000 RPM in response to the controller activating the deterrence device.

13. A back up system for ride-on equipment comprising: a sensor configured to sense an object within a back up region of the ride-on equipment; a deterrence device configured to emit a sound; an indicator; and a controller configured to: determine whether the sensed object is within the back up region based on feedback from the time of flight sensor, and activate, in response to determining that the object is within the back up region, the deterrence device and the indicator.

14. The back up system of claim 13, wherein the deterrence device includes a siren having a rotatable fan and a stator surrounding the fan, the stator including a first plurality of fins and the fan including a second plurality of fins.

15. The back up system of claim 13, wherein the sensor includes at least one selected from a group consisting of: a millimeter wave sensor, an ultrasonic sensor, an infrared sensor, and a camera.

16. The back up system of claim 13, wherein the ride-on equipment is a lawn mower including operating handles configured to control forward and reverse movement of the lawn mower, and wherein the indicator is located on the operating handles.

17. A back up system for ride-on equipment comprising: a plurality of back up sensors configured to sense an object within a back up region of the ride-on equipment, the plurality of back up sensors including at least one camera; a deterrence device configured to emit sound waves in a frequency range from 17,000 Hz to 19,000 Hz; and a controller configured to: determine whether the sensed object is within the back up region based on feedback from the plurality of back up sensors, determine whether the sensed object is a living being based on feedback from the camera. activate, in response to determining that the object is within the back up region and that the object is a living being, the deterrence device.

18. The back up system of claim 17, wherein the controller is configured to activate an indicator to indicate that a living being is within the back up region in response to determining that the sensed object is a living being.

19. The back up system of claim 17, wherein the deterrence device includes a siren having a rotatable fan and a stator surrounding the fan, the stator including a first plurality of fins and the fan including a second plurality of fins.

20. The back up system of claim 19, wherein the first plurality of fins and the second plurality of fins include the same number of fins.
